

Pacific Biomedical Network Meeting

PACIFIC BIOMEDICAL ENGINEERING NETWORK (PBEN) MEETING REPORT

29 – 31 May 2023 Tanoa International Hotel, Nadi, Fiji

Prepared by the Pacific Community

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Meeting papers and presentations are available at <https://phd.spc.int/pbn-meeting-documents>.
PBEN recommendations are in [Annex 1](#). The list of participants is in [Annex 3](#).

Meeting Objectives:

1. The objectives of the meeting are for member countries to share Biomedical experiences, challenges and to strengthen partnerships towards addressing Biomedical priorities in Pacific Island Countries and Territories (PICTs). Topics of interest include:
 - Capacity building: Strengthen knowledge and skills through training. The 1-day certification workshop will be on Biomedical Electrical Safety Testing training.
 - Leadership: Strengthen leadership and management skills for qualified biomedical technicians targeted towards country's needs.
 - Network: Support biomedical services and skills through strong collaboration and partnerships.

Opening

2. Opening remarks were provided by Dr Silina Motofaga, Team Leader of the Clinical Services Team, on behalf of the Director of the Public Health Division Dr Berlin Kafoa. The opening prayer was given by Mr Nelson Vaguna from Tuvalu. Dr Motofaga then welcomed participants and provided a historical overview of meetings of the network to-date, from the first workshop held in Nadi in 2014, training held in 2016, and the most recent 2019 meeting that was held to discuss progress in the region since 2012. She also noted that a situational analysis was conducted in 2016/17, that will be updated this year with the assistance of data provided by the participating PICTs. She emphasised how the meeting serves as a forum to table issues and recommendations nationally and for the region and reported to the Director of Clinical Services (DCS) Meeting, Pacific Heads of Health and Pacific Ministers of Health meetings respectively. The objectives of the meeting were highlighted, and in particular capacity building in the areas of workforce and education, leadership and mentoring, and strengthening of the network. She also encouraged participants to share ideas throughout the meeting, formally and informally, and make use of the opportunities to connect with colleagues across the region.

2.0 Leadership and Administration

Session 2.1 Biomedical Leadership and Administrative Responsibilities

Presented by: Ms Helen Sinclair, Procurement Manager, Rarotonga Hospital, Cook Islands

3. Te Marae Ora Hospital has a staff of 44, currently with no technical biomedical staff. The procurement manager maintains a functional relationship with biomedical expertise and the finance and procurement team for the management of biomedical services.
4. Key challenges experienced for biomedical services include:

- Limited budget allocated for asset replacement, servicing and parts (biomedical is part of hospital health services, and thus are required to prioritise funding of fixed and expense assets every year in collaboration with other services including clinical services and primary health care).
 - Supply chain issues due to a limited supplier network and lack of purchasing power. There are also ongoing challenges due to delivery timeframes enhanced by limited space availability on freight forwarders.
 - Lack of Human Resources. The recruitment of 2 currently biomedical positions is a priority for the Cook Islands as currently only 1 locum staff member in place from Samoa. Expressions of interest have been received from Samoa and the Philippines, but the team are actively seeking expressions of interest for the role.
 - COVID-19 has had a number of impacts on biomedical services including a shift in demands for care, and increased supplier costs.
5. Key achievements for support to biomedical services include centralised procurement processes that are now coordinated by the finance team allowing for more efficient funding availability and cost-effective procurement. New processes undertaken during COVID-19 including direct liaison with manufacturers rather than distributors (e.g. Sturdy, SinoHero, Edan, Nidek, Fisher & Paykell, Phillips, Stryker, Airsep), and increased networking capacity has led to more cost-effective procurement of quality medical equipment and improved inventory and supply chain management. COVID-19 also allowed for the set-up of Negative Pressure Wards, PCR Lab, CT scan and O2 Plant upgrades. Throughout these activities the team was able to continue to maintain risk management processes of medical equipment in accordance with health and safety requirements
6. **Recommendations:**
- i. Increase funding to effectively maintain existing service contracts, replacement parts and equipment.
 - ii. Funding support required for capacity building of staff to support specialised work.
 - iii. Funding support required for human resources – e.g.: locum services for shortage of staff

Session 2.2 Medical Equipment Management: Procurement of Accessories and Spare Parts

Presented by: Ms Virisila Livicala, National Biomedical Coordinator, Fiji Pharmaceutical and Biomedical Services Centre (FPBS), Suva, Fiji

- 7. The Ministry of Health and Medical Services (MHMS) has over 220 facilities across Fiji at the tertiary, divisional (A, B, C), and nursing station level. There are currently 24 biomedical staff based cross CWM, Western, FPBS and Labasa.
- 8. There is a need to look at how to strategically approach procurement and management of biomedical equipment within given timeframes, including engagement with stakeholders. There are 3 main activities required in the strategic procurement of biomedical equipment: annual audit; purchase and replacement plan for equipment; and addressing HR/workforce needs. Other activities for biomedical procurement and management include routine maintenance; corrective maintenance; expansion of services (due to increase in population size and need for increased

services); and new services (universal health coverage has decentralised health services to PHC facilities).

9. The tendering process for procurement of equipment can be lengthy and procedures must be followed. To facilitate the complete process from tender to receipt of items takes 6 months. If smaller-scale procurement <\$50,000, a request for quotation can be sought which is a shorter process that takes 1-2 months. Procurement thus requires adequate planning and dialogue with clinical service networks including biomedical for informed procurement. Finalisation of specifications, as well as quantification of spare parts based on increases in equipment is essential to ensure adequate supplies. Selection of equipment happens during the evaluation process. Tendering and requests for quotation occur following this planning process.
10. Key challenges experienced by the Ministry in the procurement process include:
 - Contracting can become a bottle-neck – different stakeholder/Ministry that handles the approval of contracts, need for in-house senior legal officer to streamline contracting processes.
 - Delayed negotiations for price variation on contracts – spare parts are becoming increasingly expensive, need to negotiate a fixed-rate cost to negate variations and ongoing impacts of COVID-19 pandemic.
 - Fully utilising funds is difficult due to lengthy procurement and tender approval processes – need for advance planning and early tendering to address this issue.
11. Key developments in the Ministry for improvement of procurement has focused on the utilisation of budget. Budgets are assessed annually to track progress on implementation and allow planning to monitor and identify risks and responsibilities. Additionally, working with donors has allowed the team to utilise different procurement processes in a more timely manner to ensure that services are restored quickly when equipment is not functioning correctly.

Session 2.3: Improvement Initiatives for the Biomedical Engineering Team at Colonial War Memorial (CWM) Hospital

Presented by: Ms Vasiti Taylor and Mr Mikaele Tikoesuva, Fiji Program Support Facility, Hospital Supply Chain Support, Fiji

12. As of January 2023, the Colonial War Memorial (CWM) Hospital biomedical engineering (BME) department is made up to a total of 22 established staff, as well as three project officers and 11 FNU trainee students. The Department is implementing a supply chain management (SCM) improvement initiatives reform project, supported by the Fiji Program Support Facility (FPSF) to enhance the availability, reliability, and storage of biomedical equipment for patient care. CWMH is 527 bed hospital and acts as a referral hospital for 2 major universities. Supply chain reforms for essential medicines and medical equipment is a key pillar of FPSF program, linked to the national interest and vision of MHMS for a healthy population.
13. FPSF support has also focused on leadership, capacity-building and networks. Activities that are being supported include coaching; appointment of improvements officer; ICT capacity building including contract with Beyond Essentials for use of Tupaia; and additional storage for new biomedical equipment. The team is with MHMS on procurement policies and procedures.

14. Obsolete equipment is a key bottleneck at CWMH. A Board of Survey for obsolete biomedical equipment for decommissioning has been completed and identified >1.65 million FJD in obsolete equipment. Further reviews of equipment lists should be clinically driven, working together with clinicians on planning and needs. Disposal of obsolete equipment requires further work to ensure adherence to environmental standards and reduction of carbon footprint.
15. The 5S Project, based on 5S Kaizen principles, commenced in Feb 2023 to improve the workshop and storeroom and enhance asset management. Phase 1 (sorting) and Phase 2 (setting in order) have been completed whilst Phase 3 (shine - cleaning and painting of workshop) is still in progress. Once complete, there will be a focus on continuous improvements via regular audits and staff training to sustain and standardize the 5S practices.
16. Several areas of opportunity have also been identified for further work and engagement including:
 - Development of a cadre of biomedical professionals via a job evaluation exercise, pathways and certification. Upcoming support to be provided includes a specialist review of BME systems and workforce, as well as supporting BME workforce to have formal training opportunities such as Australia Awards Scholarships.
 - Good governance under the Act, such as improving efficiencies across committees for greater collaboration including the review or merging into a National Committee. Concurrently there is a need for national standards, a clinically driven BME list and improved asset management.
 - A strategy for sustainability, considering outsourcing and regionalism and how to move risk to providers of equipment.
17. The key challenges faced in implementing Quality Improvement Programs (QIP) in the BME department have been:
 - Coordinating improvement initiatives while balancing essential services such as preventative maintenance, particularly when supporting remote locations.
 - Capacity to conduct market research for evaluating and making decisions on equipment specifications for spare parts and procurement of new equipment.
 - Effective communication and coordination among various internal units of the hospital with different organisational structures including clinicians, administrators, nursing unit managers, hospital services unit, asset management unit, digital health team and various external vendors.
 - Budget limitations and human resource constraints continue to pose consistent significant challenges to improvement initiatives such as training and professional development programs.

18. Recommendations for National Government:

- i. Undertake a review/job evaluation exercise for specialist biomedical engineering roles including structuring pathways for relevance and towards proper people and performance systems.
- ii. Support people and performance through Continuous Professional Development (CPD) including facilitating problem-based learning during department meetings or otherwise, to promote continuous learning.
- iii. Implementation of mSupply and Tupaia including training and certification of the BME officers on these software management systems.

- iv. Biomedical asset and inventory management should include assimilation of the existing assets registry into the new management system to meet regulatory standards. Regular audits should be conducted using the Tupaia registry to identify areas for improvement and enhance reporting and decision making.
- v. Improving good governance with standards and networking with other stringent regulatory bodies like FDA and TGA; reviewing current national governance structures and merge as one under the Principal Act; exploring outsourcing of high-end equipment at national and /or PHN level facilitated by key development partners.
- vi. Build operational relationships with various internal units including clinicians, to gain insights into biomedical equipment needs and challenges. Collaborate with the CWMH biomedical and clinical products committee in meetings and development strategies.
- vii. Enhance patient safety and experiences by conducting biomedical equipment electrical safety testing and validation to ensure the safety, reliability, and usability of biomedical equipment and accessories. Coordinate feedback from end-users of biomedical equipment such as patients and clinical staff, to improve the user experience and address any functionality issues.
- viii. The biomedical improvements officer to liaise with the head of department and department officers to identify improvement initiatives for Monitoring, Evaluation and Learning including production of monthly progress reports to track performance and identify areas of potential investment and further improvement.

19. Regional Recommendations:

- i. Proactively and intentionally participate in the Biomedical Pacific Health Network for capacity development opportunities as part of standardization and regionalism.
- ii. Solicit support at conferences, workshops, and training programs to stay informed on the latest updates within the biomedical engineering community, both local and regional.
- iii. Work with SPC, WHO and BPHN executives and member countries to develop global, regional, and national mutually agreed impact, outcome, and output-based indicators to measure effectiveness and for lessons learned through QIP cycle.

Session 2.4 Questions and Answers; Group Discussion

- 20. Majority of country representatives/biomedical staff present at the meeting indicated they also have responsibility for the procurement of biomedical equipment and supplies within their facility.
- 21. American Samoa noted challenges to procurement timelines and budget were common.
- 22. Papua New Guinea provided an overview of reforms of the procurement system at the national level including the establishment of a National Procurement Authority so that procurement can take place within the Department of Health. Committees were put in place in the Department to streamline decision-making. They noted that planning is important, however priorities must shift based on the budget made available and the needs.
- 23. Fiji indicated that when donor partners support with equipment, a donation policy in place where there is a checklist that is used to collaborate on supply of standardised clinical products

- (specifications are provided including compatibility with existing equipment and consumables). Standardisation of equipment is a challenge and agreements should be made with donors to ensure that they will provide supply spare parts for the equipment procured for its lifespan.
24. Tuvalu acknowledged that they have a much smaller budget and plan for spare parts only rather than new equipment for which they work with the donor agencies such as DFAT and WHO.
 25. Tokelau stated that they seek the advice of a biomedical adviser for types of equipment and procurement processes. Often need to procure new machines when spare parts are not able to be obtained for existing equipment. They noted that though procurement can be a lengthy process, there is budget within the department.
 26. American Samoa emphasised the need to request for preventative maintenance schedules of used medical equipment that is donated PICTs.
 27. Papua New Guinea added that donations in the country must comply with policy requirements. Management of asset life cycle is crucial, and learnings should be taken into the procurement process to decrease down-time of critical equipment that is required in health facilities.
 28. Fiji explained that critical equipment (e.g., in the laboratory), is covered under preventative maintenance and comprehensive service contracts for routine and corrective maintenance. Both MRI and CT machines are monitored remotely by contracted suppliers, who have a responsibility to ensure minimum down-time. When selection of suppliers is made there is focus on local support services to ensure services are available quickly in-country. It is important to consider tendering for spare parts for the lifespan of equipment from the supplier as well as securing budget from the Ministry of Health at the time of initial procurement to ensure sustainable use of equipment into the future.
 29. Fiji also explained that shorter time frames are possible for urgent equipment procurement (>50,000) and waivers of the tender process can occur if the Permanent Secretary and Minister support expedition based on specifications and emergency criteria. Justification of the case is required by evidence of impact on clinical services, including statistics.
 30. Cook Islands requested further information from other participating countries regarding the disposal of old or obsolete equipment.
 31. Fiji acknowledged that they work closely with the assets department, and the approval for disposal of equipment must be gained from Ministry of Finance. Disposal is completed directly through Ministry of Environment in accordance with the Environmental Management Act, which includes biomedical waste including equipment. Currently 2 vendors available in Fiji that are certified to manage disposal of this equipment. An area that requires a lot of attention, as well as consideration of carbon footprint and environmental impact and proper waste disposal.
 32. The WHO representative proposed to liaise with the South Pacific Regional Environment Programme (SPREP) which looks at biomedical waste and biomedical equipment disposal. They also emphasised the need to address the issue with both donor and recipient countries and invest and cooperate with agencies like SPREP to guide processes.
 33. Fiji noted that their MHMS has a coordinating committee for development partners and that there is a need to promote equipment requirements and donation policies in this forum as they are often receiving equipment that is at the end of its life cycle. User discipline as another issue that needs to be addressed for increased lifespan and sustainability of equipment.
 34. Cook Islands explained that for the disposal of equipment a local company removes copper and stainless steel from equipment which is sold on to New Zealand. Funds obtained from this process are then used to procure new equipment. They promoted this a method that could be investigated by other PICTs.

35. Tuvalu requested further information from Fiji regarding the length and number of student attachments in the Biomedical Department. Fiji confirmed placements were for a period of 12 months with a maximum of 15 students at a time. It was noted that bilateral discussions would be required to open this opportunity to other PICTs.
36. Papua New Guinea reinforced that registration of technicians is critical, as is the need for further capacity building and development of career pathways. It was noted that they also find the disposal of biomedical waste challenging, particularly radiology equipment.
37. American Samoa noted that around 40-50% of equipment servicing (including disposal) is outsourced, particularly for specialist equipment for operating theatres, laboratory and OBGYN services. They also provided an example of finding solutions to biomedical waste disposal when incinerators were not possible due to environmental legislation – they now have Rotoclaves where old equipment is sterilised and then grinded before being transported to landfill.
38. Fiji confirmed that following the completion of the Board of Survey, the identification of obsolete equipment was to be addressed by replacement plan.
39. The WHO representative explained how European laws are in place that require suppliers and manufacturers of medical equipment to be responsible for the management and disposal of obsolete medical equipment/waste, and strongly encouraged the need for PICTs to look at the option of developing similar legal requirements and frameworks.
40. The representative of Device Technologies noted that there is an expectation to incorporate provision of parts/supplies at Point of Sale but that this should be balanced with issues of stockpiled equipment that can create issues around expiration of warranty. They also provided the example of 'Humpty Dumpty' as NGO that had a policy regarding the donation of medical equipment for neonatal care. They promoted the need for countries to create similar lists of requirements for donations, and that these organisations could provide a good example model for donation policy.
41. Samoa confirmed that they are also attempting to better enforce their donation policy to ensure compliance from donors. They asked Fiji for further information about how the Tupaia System will be used for the registration of asset and service records. In response, the FPSF Program noted that it will act as an extension of the mSupply system and that Tupaia will commence work on the system in June/July 2023.
42. The Biomedical Consultant for Tonga emphasised the need for engagement and support from political leaders to ensure that donation policies and procedures are followed, and procurement and disposal aligns to needs of health ministries. Donations should be coordinated by the biomedical department and there is a need for enforcement of this.

3.0 Health Emergencies – COVID-19

Session 3.1 Oxygen Scale-Up Initiative in the Pacific Region

Presented by: Stefan Adriaenssens, Regional Focal Point for Operational Support and Logistic, World Health Organisation (WHO)

43. Two important international developments in the global context currently are the inclusion of medical oxygen in The International Pharmacopoeia, and most recently, the adoption of a resolution on increased access to oxygen recently at the World Health Assembly. This will influence and enhance work for oxygen capacity development in the region.

44. The oxygen scale-up initiative commenced in 2020 and so far, has overall progress at 85% (includes implementation of wider range of activities outside PICTs). 14 Pressure Swing Adsorption (PSA) Oxygen Generators were procured for the region; the equipment is provided in shipping container units which include: generator, compressors, buffer tanks, control system, filters and an O₂ analyser. The containers are insulated, with high saline and humidity protection to try to improve lifespan/sustainability of the units which are projected to be 25 years. Most units include a filling station with provision of cylinders. Spare parts are provided for 3 years of continuous operation.
45. Installation has occurred in Fiji, Tonga, Vanuatu, Solomon Islands with 3-4 days of commissioning and BMET refresher training provided as part of installation.

46. Recommendations:

- i. Work with the relevant authorities to adopt and implement standards and regulations governing medicinal oxygen while promoting third-party quality assurance.
- ii. Support the development of sustainable energy solutions, including the use of renewable energy sources, for the PSA oxygen system.
- iii. Continued capacity building & training of biomedical technicians and engineers.
- iv. Implementation of cost-efficient PSA oxygen plant operations.
- v. Creation of a regional platform for knowledge sharing and capacity building.

Session 3.2 The Fiji Oxygen Project

Presented by: Senimili Vadei, Biomedical Engineer, CureKids

47. The Fiji Oxygen Project (FiO₂) stems from research conducted in 2014 regarding pneumonia deaths in children. In 2015/16 a pilot project commenced at Nausori Health Centre to install oxygen concentrators and a solar system to support power availability. From 2017-19 assessments were undertaken to compare the use of concentrators and cylinders for both cost and mobility. At the same time, a Steering Committee was formed for governance of ongoing decision-making for the project. The project continues up until today with funding from MFAT, DFAT and Rotary.
48. Achievements to-date include:
- Coverage of 44% of health facilities with oxygen concentrators. Over 250 concentrators have been installed, and 100 pulse oximeters and other equipment distributed including regulators and consumables.
 - Development of pathway/SOP/guidance for domiciliary oxygen therapy at home in consultation with clinician team, currently being provided at no cost.
 - 12 solar systems installed in health facilities that were without reliable and stable power supply, throughout Fiji. The project is able to provide support with own procurement and purchasing processes that lessens burden on health system and allowed for timely delivery of equipment. In several locations, the supply of solar power was extended to government staff housing to try to improve staff conditions and retention. The systems can be monitored online

remotely for ease of operations and maintenance and the team continue to provide ongoing technical assessments and support.

- Training provided to 1500 clinical staff in basic utilisation and maintenance for the concentrators that reach 50,00 patients. Training has also been conducted on cyclone-readiness SOPs and basic tools for solar power systems. Online and in-service training for 20 biomedical technicians was provided on concentrators and the solar system.
- Collection of routine maintenance data and training data that will assist in preparation of a technical paper to describe experiences with oxygen and the sustainability of oxygen concentrators and other supply methods in Fiji.

49. Challenges in the implementation of the project include:

- Accessibility and logistics including transportation when working with health facilities in remote areas.
- Workshop space and storage of equipment which has been further compounded by the number and variety of donated equipment.
- Gaps in capacity and knowledge across some key areas.
- Reliable equipment management and tracking that allows for better monitoring, usage and distribution of medical equipment to the health facilities around the country.

50. **Recommendations:**

- i. Integration of CureKids project personnel into MHMS.
- ii. Adoption and operationalization of the National Oxygen Policy so that the biomedical team will have the mandate and support in place to continue adherence to the vision that no person should die due to lack of oxygen in Fiji.
- iii. FiO2 Steering Committee oversight of ongoing work related to improved access to oxygen, chaired by Dr Luke Nasedra and Dr Eric Rafai (MHMS).
- iv. Research outputs and dissemination which will assist in decision to understand most effective and efficient equipment, and the experience of community, health and project staff, in order to extract proposed solutions for improved access to oxygen, including synthesising feedback on the National Oxygen Policy.

Session 3.3 Device Technologies

Presented by: Mr Nicholas Arnold, Device Technologies

51. The representative shared his experience of work completed in Samoa 2019 and the challenges faced. The number of donors for facilities, equipment and consumables led to significant challenges with compatibility across different departments and for patient transport, highlighting the need to ensure specifications are provided to donors and adhered to.
52. In 2021 Device technologies worked with SPC and UNICEF for the supply of ventilators and humidifiers supplied to 11 PICTs. Significant challenges were faced due to COVID-19 and not being able to being provided face-2-face installation and training so the team shift to live-streaming assistance for installation and training. The company also created online resources and channels for support and information that will continue to benefit/support countries across geographical challenges.

53. There is a significant need for, and benefit to, ongoing training and support provided by suppliers.
54. PICT should consider/assess reusable consumables for reduction in both costs and disposal challenges.

Session 3.4 Questions and Answers; Group Discussion

55. The FPSF noted the common challenges regarding professional development opportunities across PICTs and suggested need for regionalisation of training, education and capacity-building including development of a regional plan.
56. Fiji flagged that engagement of clinicians on the importance of following donation policies is required to create sustainable procurement plans, and awareness of the significant costs associated with not following the plan.
57. Kiribati commented that endorsement of Donation Policies at higher levels of hospital and Ministry governance has been hard to obtain for years.
58. Solomon Islands noted that their donation policy is currently in draft stage.
59. Samoa explained how the equipment committee is responsible for drafting policy, which is then sent to hospital management for endorsement and implementation.
60. Tuvalu noted that the price for oxygen cylinders has increased significantly due to COVID-19 and requested information from the group on their experience of procurement for medical gasses/air, including the benefits and challenges associated with the use of cylinders or plants.
61. American Samoa shared their experience with COVID-19 and the need for ventilators. Procurement specified built-in compressors as no medical air generator and a lack of supply of cylinders. They also queried as to whether other countries, including Fiji, had any regulations for certification of systems installation and usage of medical gas systems. Fiji noted that their sole supplier, Oceania Gas, has responsibility for installation of piping system and provision of cylinders and are responsible for certification of this equipment.
62. CureKids confirmed that the transition of existing staff including technicians based in the Biomedical Department are to be absorbed into current MHMS structure and budget. Though transitioning staff, CureKids will continue to support training, logistics, and HR as well as working with steering committee. They hope that the transition will allow them to expand their work to other PICTs.
63. SPC's Clinical Services Team Leader emphasised that capacity-building and on-the ground training is important. Short-term support is available to countries who have limited capacity.
64. SPC's Clinical Services Adviser noted some of the challenges of oxygen concentrators including transportation of patients and the need for high-flow concentrations of oxygen.
65. CureKids confirmed that they have procured portable resuscitation kits (Oxy-Viva) rather battery-operated concentrators to assist with transportation of patients. The support provided by the organisation for the supply of concentrators can help redirect MHMS financial and procurement resources to other BME needs.
66. Papua New Guinea noted that they have benefited from the assistance of USAID and UNICEF Australia to complete an oxygen project across the country to reduce paediatric pneumonia. Solar power was also used in their context. Noted that for both PSA plants and cryogenic plants regulatory compliance is critical, aligned to therapeutic goods DGA/TA requirements. Capital investment for the 2 systems is required.

67. Palau currently has an oxygen plant in place and have ongoing support from multiple countries including the United States, Taiwan, Japan, and Australia.
68. Most participants indicated that they are responsible for servicing needs in their countries and have received training on the required equipment. There was a mixed response to availability of parts for the equipment (Cook Islands indicated yes; Tuvalu indicated they do not).

4.0 Capacity Building and Ongoing Professional Training

Session 4.1 Fiji National University Certificate IV for Biomed

Presented by: Assistant Professor Ronesh Sharma, Electronics Department, School of Electrical and Electronics Engineering, College of Engineering, Science & Technology, Fiji National University (FNU)

69. Fiji National University (FNU) Overview: 5 Colleges; 2 Centres; TVET and higher education, certificates through to doctorates; Programmes registered through FHEC, ENZ accredited on selected Diploma programmes.
70. Programme Structure Cert IV Biomedical Engineering, delivered under the School of Electrical and Electronics Engineering. Includes a 12-month attachment (submission of log-books) to be eligible for graduation. Entry requirements are a Pass in Year 12 Certificate Examination, with pass in English. Course is delivered at the Samabula Campus, Suva over 4 quarters (6 subjects/modules per quarter but not all are examinable).
71. The program was originally developed in 2016 and introduced, since this time it has been difficult to obtain biomedical staff, so enrolments were limited; the program restarted in 2019 with the recruitment of 2 x international staff. Since then, the numbers throughout 2020-21 have been impacted by COVID-19 but have grown from 4 enrolments in 2019, to 24 in 2023. 5 students graduated from the program in 2022.
72. Other key progress and achievement have been:
 - Dedicated biomedical lab and upgrade of lab equipment – previously only had electronic lab, currently developing a biomedical lab with specific equipment.
 - Enrolment of new, qualified staff members for delivery of program.
 - Review of program with feedback from industry.
73. Key challenges in the delivery of the program have been recruitment and retention of appropriately qualified staff including higher postgraduate staff and diploma level trainers. Allocation of budget for these staff has also been challenging. Set-up of the service centre hub and development of the Diploma level program have also been delayed (and impacted by COVID-19).
74. **Recommendations:**
 - i. Recruitment of experienced and Masters/PhD level biomedical staff and updated biomedical trainers essential for program delivery.
 - ii. Continue to prioritise the setup of biomedical service center hub.
 - iii. Review of certificate Programme with industry feedback to improve training and outcomes.
 - iv. Development of Diploma level Programme an ongoing priority.

Session 4.2 Continuity of Local Biomedical Expertise

Presented by: Ms Catherine Fiu, Principal Biomedical Officer, Samoa

75. The Biomedical Engineering Department is responsible for over 6000 medical equipment in 2 main hospitals and 10 district hospitals throughout Samoa. The team consists of 7 staff members, with multidisciplinary backgrounds and expertise in biomedical science and mechanical engineering. They are responsible for the management of the full lifecycle of biomedical equipment: procurement – technical specification, evaluation, bidding; acceptance and commissioning; maintenance, support, and user training; and disposal and replacement.
76. Importance of systems and protocols so that staff have adequate skills and knowledge, to maintain and improve quality of service, and strengthen independence and sustainability (including independence from overseas expats).
77. Documentation that is currently in place to support biomed activities include:
 - Biomedical engineering procedure manual – guidelines followed on day-to-day basis. Complements other national policies including the National Medical Equipment Policy; AS/NZ:3551 (2012) Standard (Management Programs for Medical Equipment); TOR for Medical Equipment Management Committee; and Donation Policy. Explains the full scope of BME services in detail; also outlines organisational structure; Occupational Health and Safety (OHS); and supporting documents.
 - The Asset Record contains all the of the medical equipment under biomed care and includes all information about medical equipment pieces (i.e. manufacturer, model, installation date, warranty date, cost, availability of user and service manuals etc.).
 - The Service Record records all work performed, from administration tasks, acceptance of new equipment to maintenance and training. It is useful for when technicians can look back at repeated faults for assistance.
78. Key challenges have been:
 - Influx of medical equipment – measles and COVID-19 led to a large amount of testing and acceptance of equipment, complemented by supply of consumables. Eg. 260 oxygen concentrators were received. Recording of this equipment has been time-consuming (both inputs and upload to online system).
 - Training opportunities – there are currently inefficient methods of recording and a lack of formal training opportunities to upskill BME staff.
79. Key Opportunities and Development has included:
 - Continuous Professional Development (CPD): Delivered fortnightly, led by the BME team. Presentations and/or technical training on a pre-selected piece of medical equipment. Includes mechanism of operation, preventative and user maintenance. Quizzes are carried out every 2 months to test understanding and further solidify skills.
 - Customised forms – including Acceptance and Preventative Maintenance Forms. Improves quality of work done on an equipment as the tests are conducted according to manufacturer advice and improves efficiency when performing acceptance tests and preventative maintenance.
 - Online Installation Training from Manufacturers - Allows local staff to ask specific questions to trainers. Convenient as this type of training can be done at any location at any given time.

Builds professional relationships with trainers and manufacturers to make repair and procurement processes easier.

- Renewal of Medical Equipment Management Committee (MEMC) including TOR. MEMC determines the Priority Equipment List and distributes it to respective donors, expedites medical equipment procurement and enforces the Donation Policy.

80. Recommendations for National Government:

- i. Up-skilling BME staff by attending technical trainings (either on-site, on-line or at manufacturer location).
- ii. Implement a new asset and service registry system that is more efficient than existing one, together with the appropriate tools: e.g. Tupaia System.
- iii. Regular updating of procedure manual, and related supporting documents for the sustainability of BME service.

Session 4.3 Challenges of Human Resources

Presented by: Ms Kulakimata Vaha, Biomedical Technician Trainee, Niue

81. The organisational structure for the Department of Health is currently understaffed with a total of 51 across all sectors. These human resourcing challenges are present in the Biomedical Department - there is currently no qualified biomedical technician or Engineer in Niue, and assistance consistently needs to be sought from overseas though availability has not been consistent. The lack of staff has impacted on the ability to install and commission new medical equipment, particularly during the COVID-19 pandemic. It has also resulted in a lack of mentoring and training opportunities for the current trainee. The development of local staff is needed for long-term sustainability.

82. Strengths and opportunities currently have stemmed from training opportunities and established networks with other BMEs for have ongoing support. New systems have been installed and existing systems repaired utilising external expertise both on-line and face-to-face. Removal of obsolete equipment has also commenced. Key challenges for the delivery of these activities have included the cost and availability of international specialists, as well as time differences, technology and network capacity for online support.

83. Recommendations:

- i. Niue Foou Hospital to procure basic but essential medical test equipment (e.g. Multimeters) to use during installation and preventative maintenance work.
- ii. Niue Foou Hospital to provide continuous professional development (CPD) for biomed technician trainee to further develop technical knowledge and skills.
- iii. Exchange partnerships between PICTs, including between Niue and Samoa, in order for trainees to experience the workflow and overall setup of the biomed unit.

Session 4.4 Professional Development

Presented by: Mr Nelson Vaguna, Biomedical Technician and Mr Semisi Finau, Biomedical Engineer Consultant, Tuvalu.

84. Professional development is broad term used to describe a range of activities. For the Tuvalu workforce this has included: gaining new skills and knowledge, higher education (formal qualifications), capacity building, upgrading tools and equipment, Occupational Health and Safety in workplace and registering to Professional engineering license.
85. Some of the key achievements to-date include:
- Receipt and testing of donated equipment from COVID-19 pandemic.
 - Technical Adviser introduced software for electrical safety testing.
 - Training of health care professionals on use and maintenance of biomedical equipment that created greater ownership and responsibility amongst staff for equipment.
 - Delivered sessions on career pathways for biomedical roles to High School Students to promote science and engineering (biomed, lab, radiology etc.) as a career pathway in Tuvalu.
 - Promotion of improved practices for maintenance and care of equipment by establishing 'Parking Area's and developing coverage of equipment for protection from environment.
 - Development of an Asset Registry with unique identifiers to ensure tracking and responsibility. The registry has also allowed for preventative maintenance and performance verification of equipment.
 - Development of policy making and managerial skills as well as documentation and reporting templates.
 - New Projects in development include a CT Room, oxygen plant location and preparations.
 - Exploration of further study, attachment, training and mentoring opportunities.

Session 4.5 Questions and Answers; Group Discussion

86. Samoa queried as to whether the placement component of the Certificate IV needed to take place in Fiji. FNU confirmed that the placement does not need to be in Fiji and can be in home countries.
87. DFAT noted the importance of establishing biomedical training programs that balance both practical and theoretical components. Also queried the recognition of the Certificate IV by other universities regionally. FNU confirmed that recognition had not yet been established with other universities.
88. Fiji queried as to whether workforce assessment had been done and what efforts had been made to ensure placement and job-availability following the graduation of students. FNU noted assessments had been made but no formal assessment available.
89. SPC queried the availability of alternative methods of entry to Certificate IV and if bridging options were available for those who have not completed Grade 12. FNU confirmed that science bridging courses are available and that on a case-by-case basis assessment of continuous experience could be made for entry to the program.
90. FPSF queried to FNU and others as to what support could be provided to Health Facilities that are hosting placements for students for infrastructure and support for the students during their time on attachment.

91. Tuvalu queried as to the appropriate role/level of biomedical staff aligned to the certificate program available. FNU confirmed that the Certificate Level IV was equivalent to an assistant biomedical technician and that a Diploma level qualification would be a technician.
92. Samoa queried as to what scholarships can be made available to staff for study. FNU confirmed no scholarships currently in place, to be negotiated within national governments and with partners.
93. American Samoan noted the need to ensure/recognise educational pathways in the region for those who have US qualifications.
94. Papua New Guinea spoke on reforms which took place to establish National Industry Standards for BME/Clinical Services. They also provided background on their own Biomedical Engineering program which in 2020 had its first intake at PNG University of Technology. The University worked with US/India to structure the programme and gained approval for the course through the Qualification Framework and University Board. It is currently a 4-year Bachelor Programme under the Applied Physics Department at Engineering School. From the first cohort of 15 students, 7 students are on track to graduate at the end of the year. There are currently 23 hospitals in the country and growing demand for this type of expertise. They are working to ensure an industrial training pathway (including placements) is properly developed to ensure applied competencies are developed in students. A residency period for 2 years is also being trialled with clinicians and the University to improve career pathways and employability for graduates. Development of industry within the country critical to housing these graduates beyond the Ministry – maintenance, supply and installation of equipment by private sector all currently in place. All work is guided by standard frameworks between the University, Government, Ministry etc. MoUs have been made and work is being done closely with specialist areas – lab, radiology etc.
95. Samoa confirmed that the adapted forms noted in their presentation are not currently officially endorsed at the government/ministry level and that these are still paper based rather than electronic.
96. Papua New Guinea commended the hard work and passion from young female biomed in the region and noted that they too are hoping to better develop gender equity in their own biomedical workforce which has been male dominated.

5.0 Situational Analysis

Session 5.2 Regional Biomedical Services – Situational Analysis

Presented by: Ms Sunema Talapusi, Biomedical Adviser, Clinical Services Programme, SPC

97. A previous Biomedical Situational Analysis survey was conducted by SPC in 2014 and 2019, with an additional survey conducted in 2022 to identify key challenges experienced by the biomedical workforce and identify key areas for support. In 2023, a new situational analysis survey and request for workforce data was distributed to, and received from, 16 PICTs ahead of this 2023 regional meeting. The information collected and presented is self-reported preliminary data only and will require further collaboration with individuals PICTs to establish a complete picture of status, identify gaps, and develop future works plans.

98. Initial findings in regards the regional biomedical workforce show an increasing number of biomedical staff across the region. Comparisons with (limited) 2014 data shows an increase of 100% in total workforce numbers. There is also a much higher representation of women (increased from 8% in 2014, to 17% in 2023); and a shift in age distribution to a much younger workforce (66% of the workforce is under the age of 40). The workforce is also more highly qualified, with the number of staff with a Bachelor's, Certificate and Diploma level qualification increasing from 61% to 83% and the number of staff without any formal qualifications and training decreasing.
99. Overall, there has been gradual development of national policies and systems for biomedical services across the region, though this often has been dependent on availability and capacity of national staff. A few countries have regressed in the implementation of key components of biomedical support services and governance. Key findings from the analysis include:
- 19% of countries indicated they have 100% in-house capacity for biomedical services.
 - 24% of countries have implemented a National Medical Equipment Policy (in over half of the countries surveyed, the policy is still in the planned (24%) or drafted (35%) stage).
 - 47% of countries have national equipment committee, though half of these were reported to be currently inactive.
 - 35% of countries have implemented a biomedical procedure manual (for half of the countries (47%), manual remains at the planned or drafted stage).
100. Key challenges and issues that have been identified by respondents across multiple surveys and previous regional meetings/trainings include funding and human resources; implementation of donation policies; disposal of medical equipment; limited workspace and infrastructure; availability of essential tools and test equipment; and availability of spare parts.
101. Another key area identified as a challenge was training and ongoing support for professional development. More than half of countries who responded to the survey noted that they did not have requirements for continuous professional development and that they did not have funding for user equipment or service equipment trainings.

102. Recommendations for National Governments

- i. Verification of data and review of current policies to be undertaken with each PICT.
- ii. Endorsement to publish regional data.
- iii. Results and findings of to be shared amongst the network and distributed more widely (including research publication and dissemination in forums where appropriate and beneficial), to better understand regional and country-specific status of biomedical services and apply learnings and recommendations that address the specific needs and challenges of the region.
- iv. Support access to biomedical training leading to formal qualifications and ongoing professional development training.
- v. Consideration of retention package to provide incentives to work in the biomedical field such as higher salary, retention pay.
- vi. Award tenders only to manufacturers that provide ongoing and timely support to PICs through provision of spare parts and technical advice.

103. Recommendations for Partners

- i. Provide technical support for the biomedical workforce through the development and provision of in-person and online training in local language and access to training opportunities leading to formal qualifications.
- ii. Ensure that any biomedical equipment procured or donated meets country regulatory requirements and technical specifications, and includes provision for installation/training/maintenance, where appropriate.
- iii. Support research opportunities at national and regional level

6.0 Biomedical Policies

Session 6.1 National Medical Equipment Policy

Presented by: Mr Twaine Cecil, Biomedical Equipment Technician (BMET) Intern, Nauru

104. There is currently no National Medical Equipment Policy in Nauru and there are number of challenges to ensuring that this is progressed. Drafting requires investment of all stakeholders.
105. Key challenges currently include:
 - The Department system structure – the Secretary needs to bring the policy to the Minister for approvals, requires leadership on multiple levels. BMET part of the Health Operations Section, departmental structure can lead to challenges in approval of the policy.
 - Awareness and advocacy for the importance of the policy.
 - Limited capacity of the BMET unit.
 - Analysis of the policy and current challenges is an important step in engaging stakeholders and improving BME services. Conflicting priorities means that wider support required from wider BMET and Health Operations Section.

106. Recommendations for National Government:

- i. The working space for BMET requires further development to address current challenges for work, storage and disposal areas.

National medical equipment policy to be developed in consultation with health care services - need to consult with BMET, patients and other stakeholders including those in healthcare service operations for finalisation and implementation.

Session 6.2 National Medical Equipment Policy

Presented by: Ms Veronica Vaouli-Gasio, Chief of Engineering and Facilities (Biomedical Representative), American Samoa

107. LBJ Tropical Medicine Center is accredited as a 150-bed facility, in reality has capacity for 130 beds. The facility is semi-autonomous, the government does not have control over operations but rather appoints the board including CEO. It holds a number of accreditation types for facilities in the US including CMS (qualification for medical insurance).

108. To-date the facility has been utilising MCN Healthcare resources (library of policy for healthcare facilities – includes requirements from multiple boards) to develop tools and procedures tailored to individual facility needs. The Medical Equipment Policy is now in draft, and this regional meeting has helped to identify a number of new areas not yet considered e.g. Donation policy. The equipment committee is also currently in place but have not met in a number of years.
109. Key challenges for biomedical services include:
- Funding – It is often difficult to locate resources/funding for the purchase and maintenance of new equipment; COVID-19 has led to global shortages and specifications cannot always be met.
 - Access – Due to remote location, long lead times are required for replacement parts and/or new equipment. This equates to longer down times on critical equipment which in turn affects quality of care.
 - Training – Lack of formal training on biomedical technology, (use, maintenance and repair). Location again impacts on ability to train clinical and biomedical staff with high costs associated with sending staff to the manufacturer for training/certification.
 - The Centers for Medicare and Medicaid (CMS) conditions of participation ensures healthcare facilities are following Life Safety Codes regarding medical equipment. Proper documentation is vital to maintain compliance and there are ongoing requirements for monitoring and auditing for accreditation.
 - Obsolescence – There are no local or national standards on disposal of equipment (including environmental), and medical equipment can quickly become obsolete.
 - Contractual Services – Outsourcing means that documentation of servicing and maintenance is not required/reliable.
110. Two key areas of development and opportunity have been identified for immediate work. Firstly, implementation of a Computerized Maintenance Management Software (CMMS). Currently utilising a software program called Micromain to track work orders. Tracking of assets and work orders for maintenance and repairs and automatic reminders for maintenance will help to improve efficiencies. Secondly, a source of funding (DoI) has been identified to send biomed staff to receive manufacturer training and certification and ensure further employee development.
- 111. Recommendations for National Government:**
- i. Standardisation of biomedical equipment to ensure all equipment meets required safety standards (NFPA 99, NEC, CMS).
 - ii. Stakeholder Engagement (healthcare professional, patients and manufacturers) to ensure needs and concerns are considered.

Regulation and monitoring of medical equipment to ensure that it is safe and effective.

Session 6.3 Medical Equipment Donation Policy

Presented by: Mr Jayson Rombaoa, Director/Biomedical Engineer, Marshall Islands

112. Medical Equipment donations currently are fragmented, leading to confusion, duplication of effort, and recipients getting unwanted or incompatible medical equipment. Many different

brands and models (portable x-rays, ventilators, oxygen concentrators etc.) make it difficult for facilities to manage. Much of the equipment is unused due to these challenges.

113. It is common to receive broken or inappropriate equipment. This is coupled with a lack of training, manuals, spare parts, and consumables. This equipment is also unused requiring storage or disposal.
114. Donors know little about the impact of their donations and do not track how it is received or utilised by recipient countries. Donations are not properly planned or coordinated with the biomedical department. Marshall Islands is now looking at how to ensure medical equipment donation is coordinated in collaborated with the Biomedical Office and consideration is now being made regarding criteria for evaluating medical equipment donations.

115. Recommendations for National Government:

For inclusion in Medical Equipment Donation Policies include:

- Donors to provide a certificate of calibration if donated medical equipment is used or pre-owned.
- Medical Equipment must have a remaining life span of more than 5 years or be brand new (not obsolete).
- Donors and suppliers must provide technical training to end users of the medical equipment.

Session 6.4 Infection Prevention and Control

Presented by: Margaret Leong, IPC Adviser, Public Health Division, SPC

116. IPC as essential component in preventing/reducing Hospital Acquired Infections (surveillance, HH and IPC Plans) with significant evidence of the effectiveness of IPC programmes including hand hygiene.
117. IPC is a cross-cutting component of the healthcare system. Biomedical services are essential in ensuring that a clean, well-functioning environment and equipment is maintained as a core building block of WASH and IPC protocols in healthcare facilities.
118. Clean environment and equipment as essential for all support staff including biomedical services – there is significant risk of transmission if cleaning protocols are not followed. Biomedical staff should ensure IPC and cleaning processes incorporate manufacturer's instructions with the necessary IPC procedures. These procedures and cleaning schedules should be updated on a regular basis and staff should be trained, with responsibility assigned, for cleaning processes for equipment.

119. Recommendations:

- i. Biomedical staff should be aware of and understand the measure in place for IPC and the ABHR and HH required for equipment. Staff should ensure they are cleaning hands before attending to the machine, and before leaving the room/setting.
- ii. Providing information to clinical and cleaning staff about cleaning of medical equipment.
- iii. Include IPC in Equipment and Consumables Committee or other methods for deciding on procurement of equipment to gain their advice.

- iv. IPC Education should be provided to all Biomedical Staff on a regular basis.

Session 6.5 Medical Equipment Asset Management

Presented by: Mr Oswald Malisa, Biomedical Technician and Mr Andy Lyons, Biomedical Consultant, Vanuatu

- 120. Prior to 2022 the Ministry of Health (MoH) had few records and did not monitor medical equipment. Donation of equipment was often made directly to hospitals and wards, not coordinated by assets or biomedical staff. PICTs experience a difficult environment for equipment in both the environmental conditions and the condition of equipment upon receipt from donors. The team have worked to establish a Computerised Maintenance Management System (CMMS) to manage equipment and schedule planned maintenance. 1800 assets now registered, expected that will reach 3000 items after final audits.
- 121. Registration has allowed for forecasting of costs in the government budget annually; and scheduling and monitoring of maintenance to ensure that equipment can be available and in service for use by clinicians.
- 122. Vanuatu has implemented the FIIX system. Advantages of the cloud-based system include cost-effectiveness (no major initial purchase cost, no royalties or upgrade costs); higher level security (threat of cyber-attacks and impact of IT issues); and relatively low-cost programme maintenance that reduces cost of management of medical equipment. Additionally, the system is accessible, simple to use, and allows for scheduling of maintenance, client generated repair requests, task recording for projects and management functions such as reports for maintenance, repairs and finance.
- 123. Vanuatu is working to build up a foundational level biomedical service, there is an opportunity to establish a professional Biomedical engineering department with a need for ongoing professional development opportunities for staff.
- 124. Recommendations:**
 - i. Vanuatu MoH establishes a budget line for BME. Current forward planning for each hospital has been completed and budget submissions to maintain and repair equipment at all facilities have been made.
 - ii. Recognition is given to BME as a separate, professional discipline – currently, hospital maintenance staff are being asked to repair sophisticated equipment they have no experience, understanding or training for – as a result most equipment are discarded for minor faults that a competent biomed can easily repair.
 - iii. Relationships with biomedical staff in other countries should be encouraged as they lead to not only the professional growth and awareness of biomed across the region, but its importance in growing the experience of local biomedical staff and reducing the reliance on expensive overseas consultants and engineers.

Session 6.6 Medical Equipment Asset Management

Presented by: Mesrine Kelmal, Biomedical Technician, Palau

125. Historically there has been a lack of implementation of asset management for biomedical equipment in Palau. There is currently no CMMS that would help to streamline maintenance and work orders or recording of assets and maintenance. Additionally, there is lack of trained staff who would be able to create and maintain this system/management functions.
 126. Though it is possible to use existing solutions for Medical Equipment Asset Management despite the lack of National Policy or CMMS (Microsoft Excel, Microsoft Access, and Other Similar Software), more work is needed in creating a foundation for a solid Medical Equipment Management System. A CMMS would greatly help in Medical Equipment Asset Management by streamlining timed service intervals as well as work order requests. It would ensure that assets are well recorded.
 127. The Biomedical staff are now working on the introduction of a CMMS (FIIX) in collaboration with Australian Biomedical Engineer, Mr. Andy Lyons.
- 128. Recommendations for National Government:**
- i. Prioritise acquiring and implementation of CMMS for Palau.
 - ii. Increase staff competency in usage of CMMS in order to fully utilise and maintain the system.
 - iii. Hire appropriate number of biomedical staff to allow greater balance of work and contribution to maintenance of CMMS.

Session 6.7 Questions and Answers; Group Discussion

129. Papua New Guinea commented that capacity and planning of healthcare facilities is a common challenge across the PICTs. There is a need for standardisation reflecting capacity on the ground.
130. SPC noted that similar challenges are faced across countries regardless of size and capacity.
131. FPSF promoted the need for all to practice a One Health approach in their work. They also queried as to whether there are standardised procedures and products for cleaning of medical equipment that are included in service contracts. The group noted that equipment manuals articulate cleaning practice that ensure safety of the machine and patient. The FPSF representative also noted the importance of all professionals to be involved in IPC, and the need to include cleaning/IPC in annual budgets.
132. WHO reiterated that IPC guidelines and training resources are available online on the WHO website.
133. Device Technologies encouraged staff to work with suppliers for recommendations and guidelines for cleaning to ensure sustainability of equipment.
134. FPSF emphasised the importance of recognition for the cadre of biomedical professionals in the region and suggested this should be recommended to Ministers meeting.
135. Tuvalu noted the importance of ensuring tailored software solution for each country, relevant to their needs and systems.
136. Samoa queried the security of the FIIX system presented by the Vanuatu adviser. The adviser confirmed data is held on remote servers for enhanced security and that administrators hold authority over access for users of the system.

137. SPC recommended that a Board of Survey (BoS) be undertaken in PICTs to analyse the current status of biomedical assets, and promote to governments and donors the needs, gaps and opportunities specific to each country.

7.0 Maintenance and Ongoing Support

Session 7.1 Preventative Maintenance

Presented by: Ataata Binataake, Biomedical Engineer, Kiribati

138. Kiribati's health system is made up of 1 national referral hospital, 3 district hospitals, 34 health centres and over 70 clinics.
139. Key challenges faced by biomedical staff focus on lack of equipment and information. There are currently no BME analysers available so testing, and maintenance is dependent on visual inspection, function testing and cleaning. There is also limited knowledge on preventative maintenance and service manuals and templates/guidelines for maintenance are unavailable for much of the equipment. Access to equipment is restricted by consistent usage (lab, OT and radiology), movement of equipment within the health facility, and access to many sites is difficult (limited funding, difficulties with transport, equipment and tools difficult to transport/relocate). Staff are also overwhelmed with additional non-biomedical tasks allocated to BME staff – e.g. maintenance for equipment such as generators – which impacts on scheduled works including preventative maintenance.
140. Recent achievements include the newly developed asset register, which a TA from Australia assisted with remotely. Guidelines have been developed and implemented for use by staff, including a schedule for cleaning and preventative maintenance because of the register.
- 141. Recommendations for National Government:**
- i. BME documentation, especially the donation policy, to be revised, endorsed, and implemented.
 - ii. Provision and training on BME special analysers and development of PM standardised guideline for all equipment
 - iii. Exploring new partnerships (particularly with technical and development partners).

Session 7.2 Preventative Maintenance

Presented by: Mr Edmond Piasi, Biomedical Technician, Solomon Islands

142. Solomon Islands has 9 hospitals, located across 9 provinces. There are currently 6 biomedical staff within the country.
143. A key ongoing challenge has been training and capacity development of staff. In particular, obtaining formal qualifications (only 1 staff member has a biomedical engineering degree), but also ensuring those with formal qualifications receive hands on training and capacity development.
144. There are current gaps in leadership and governance with a lack of policy to guide the national functions of the biomedical department and limited management capacity/knowledge within the team. Other challenges faced by the biomedical department include a lack of resources –

human (trained staff), financial (there is little recognition of the Department and its needs to complete core functions) and infrastructure/supplies (working/bench and storage space, transport, spare parts).

145. There has been progress on several activities that has laid the groundwork for further development of the biomedical sector including drafting of a National Biomedical Equipment policy and development of a newly proposed structure for Biomedical Engineering Department so that biomedical support can be performed as a national level function. Additionally, there has been the installation of a new PSA plant with dedicated Biomedical staff member, to better address national medical oxygen needs. Management can now better recognise the critical importance of the Biomedical Department. Finally, ongoing refresher training has been put in place for technicians at two years intervals.

146. Recommendations for National Government:

- i. Performing national functions and development of a National Biomedical Policy as a key priority.
- ii. Restructuring of the biomedical department with a focus on recruiting new engineers/technicians to fill currently vacant positions.
- iii. Manager training to coordinate department maintenance work and maintenance management software.

Proper storage space for Biomedical Equipment separate from National Medical Stores (NMS).

Session 7.3 National Equipment Committee

Presented by: Mr Ambrose Kwaramb, Engineer/Manager Health Facilities Standards, Papua New Guinea

147. Clinical functions, and medical equipment, are important at each stage of the life cycle from paediatric through to geriatric care. PNG currently has 4,583 Health Facilities across 6 levels (from National Referral Hospital through to Aid Post). The number of facilities and equipment creates high operational costs – currently the budget allocated is not sufficient to meet the needs of the department, impacting on functions and maintenance. Governance is important for channelling issues appropriately and corporate governance (finance, HR etc.) provides important support to clinical governance. There are several levels in the health infrastructure decision making and the Medical Equipment Committee operates at the lowest level, feeding up into the Provincial Health Authority (PHA), National Department of Health (NDoH), and National Executive Council (NEC).
148. The key challenges for biomedical services span across a few key areas. First, there is currently lack of dedicated and effective medical equipment governance committees operating at hospital, provincial and national levels, contributing to lack of leadership and governance towards effective management of biomedical engineering services at healthcare facilities. Secondly, there is a lack of up-to-date data and information of medical equipment at health facilities to inform decision making, including lack of database and dashboard for effective monitoring and management of medical equipment issues. Finally, this lack of data and governance has contributed to lack of preventative maintenance, calibration, testing and

replacement or disposal of outdated or broken equipment. These issues are further impacted by inadequate funding, poor biomedical engineering facilities, and lack of capacity building/training for staff.

149. The Biomedical Sector in Papua New Guinea also has key strengths and areas under development including:

- The National Department of Health has a Biomedical Engineering section within the Health Facilities Standards Branch to manage and coordinate Biomedical Engineering Services and Standards across the 22 PHAs and PMGH and Biomedical Engineering Service capacity is available across 19 provincial specialist hospitals and national specialist referral hospital out of the 22 provinces. It is a requirement of the medical equipment management policy for all provincial health authorities to have a Medical Equipment Committee under governance charters.
- 6 main private biomedical equipment supply and maintenance companies available in PNG.
- A four-year Bachelor of Biomedical Engineering course commenced in 2020 at PNG University of Technology with the first graduates expected in early 2024.
- A National Biomedical Equipment Policy and Asset Management policy has been developed and is guiding implementation.
- Health Infrastructure and Medical Equipment Technology Standards developed and applied as part of the NHSS volume 3.
- Medical equipment replacement program annually funded through the health sector capital investment budget, with some funding for medical equipment maintenance is consistently provided under the annual budgets.

150. Recommendations for National Government:

- i. Conduct audit of biomedical equipment and workshops at PMGH and 22 Provincial Health Authorities.
- ii. Strengthen Biomedical Engineering Governance System at the National Department of Health and at 22 Provincial Health Authorities level by establishing and improving performance of committees.
- iii. Improve and strengthen the Biomedical Engineering workforce capacity and capability through capacity building training and development programs.
- iv. Improve and strengthen biomedical equipment maintenance management system at the Port Moresby General Hospital and 22 Provincial Health Authorities.
- v. Establish and strengthen system for the safe disposal of obsolete medical equipment.

Session 7.4 Support to PICTs

Presented by Mr Tociro Kitaka Biomedical Consultant, and Mr. Tuifonualava Tangatailoa, Biomedical Technician, Tonga

151. Tonga has a total population of 103,000. There are six hospitals, with the main referral hospital, Vaiola Hospital, located in Tongatapu. The location of health facilities across islands creates significant logistical challenges.

152. A key challenge for the biomedical workforce in Tonga is the set-up of the biomedical workshop. There is limited test equipment available (no electrical safety analyser) and those that are available are out of calibration and validation. Additionally, there is limited workbench and storage areas, plastic tables are currently utilised and there is no power supply (staff must use extensions cables). Staff are not provided with appropriate PPE or supplies (safety glasses, boot, clothing etc).
153. Challenges for the workforce structure include limitations in training. The highest qualification available in-country is a Trade Certificate in Electrical engineering. Currently assistants in the department have limited experience and are employed casually so are often deployed for other work in the hospital; there is also limited exposure and practical training, so they are unable to work independently creating more workload for experience technicians. The appointment of assistants as casual staff has also led to issues with lines of reporting, duties and responsibilities and sustainability. There are currently no protocols or standard methods of practice to help mitigate these issues.
154. Inventory challenges persist due to a lack of recording database (the previous database software licencing is not renewed as significant cost; there are no logbooks or records of repairs and services) and reporting templates.
155. Procurement challenges are centred around a lack of involvement of biomedical staff in purchasing recommendations (specifications and quality assessment); and the receipt of donated equipment of varying brands and models makes procurement of parts challenging. Donated equipment is also often received without consumables or information, and in some instances are not operational.
156. Progress has been made in cleaning-up of the workshop for space and storage and the development/upgrade of an equipment inventory. Additionally, staff training is taking place with a number of students training at certificate level at FNU currently. A draft biomedical structure has also been submitted for approval.
157. **Recommendations**
 - i. Continuous support for training and professional development for technicians, specifically for biomedical and report writing (documentation).
 - ii. PPEs to be provided for all biomedical staff.
 - iii. More involvement of biomedical staff in procurement processes.
 - iv. Standardisation of BME practice and documentation for the PICTs.

Session 7.5 Questions and Answers; Group Discussion

158. Tuvalu reiterated the point made by Kiribati on competing priorities and additional tasks given to Biomedical staff being an ongoing challenge for workload management.
159. Samoa also noted these challenges and explained how they have held sit-down meetings with stakeholders including hospital management to try and better demarcate roles and responsibilities for maintenance and biomedical staff which was then communicated to the broader workforce to try and address these issues.
160. Papua New Guinea echoed these sentiments and noted that the skills and qualifications of the biomedical workforce means that they need to be utilised for other urgent needs and priorities.

161. Fiji noted that having biomedical staff positioned separately from support staff (part of allied health rather than corporate services like maintenance) improved demarcation; though also noted that collaboration with support services still crucial as there is overlap on skills and needs.
162. SPC emphasised the need for detailed job descriptions for the biomedical workforce and demarcations between roles and responsibilities of different staff.
163. The Biomedical Consultant for Tonga noted the importance of management gaining high levels of understanding of the challenges faced by biomedical staff in their work.
164. Tuvalu confirmed the need to seek further support for training costs.
165. WHO emphasised the importance of completing processes upon the receipt of goods that are purchased or donated. They proposed that the Pacific Biomedical Network could be used to provide information on surplus equipment to reallocate equipment more efficiently amongst PICTs.
166. SPC requested participants reflect on lessons from COVID-19 and the need for implementing partners to work together to ensure procurement of equipment is done in an efficient and sustainable way.

Session 7.6 Remarks from the Secretariat

Presented by: Dr Berlin Kafoa, Director, Public Health Division, SPC

167. Dr Berlin again welcomed participants and recognised the importance of meeting together as a network for the first time in a number of years. He thanked everyone for their participation in discussions and involvement and expressed that SPC and other partners were looking forward to continuing to work together to support key technical development for biomedical services across the region including capacity of biomedical education and services, education opportunities, mentoring within the Pacific cohort of biomedical engineers and strengthening biomedical policies.

8.0 Group Discussions

Session 8.1 Preventative Maintenance

174. The group noted the importance of preventative maintenance to lengthen the lifespan of equipment, and in turn save money and time. They also noted it is important to track equipment history and know last service for equipment. Finally preventative maintenance helps to ensure everything is working correctly and safely.
175. The group articulated the components required for Preventative Maintenance through the acronym 'AFPS'.
 - A – Appearance. Visual inspection (IPC procedures)
 - F – Function. Buttons, displays, power cables.
 - P – Performance. Power-up and operation, changing filters and consumables.
 - S – Safety Testing. Electrical safety testing, batteries, fuses. Calibration where possible.

They noted the importance of reporting and documenting findings and the work carried out, and that the PM Sticker should be updated with the date and next dates of service.

Session 8.1 National Medical Equipment Policy - 1

176. The group outlined that the National Equipment Policy is a framework for management for equipment that raises awareness to all users and management for: definition of roles and responsibilities; ensuring equipment is safe to use; assisting with the allocation of resources; and delivering on organisational goals and objectives.
177. They noted that to make further progress on these policies, drafts need to be completed, including stakeholder engagement. Policies should be reviewed based on feedback and then submitted to medical equipment committee for approvals.
178. The group also suggested revisions to current draft generic National Equipment policies such as the inclusion of regulatory frameworks and the human resources required for implementation.

Session 8.2 Donation Policy

175. The group believed a donation policy is important for guiding Ministries of Health to ensure compliance, accountability for the donors and recipients, and sustainability of equipment.
176. They noted to progress these policies, teams need to create more awareness, possibly by taking part in the clinical services network to ensure that teams are working closely together or by writing up a circular on the policy for distribution that detailed minimum clinical standards, accountability and history, safety, and compatibility of parts and consumables.
177. For suggestions on changes or additions to current generic policies they suggested that Donors/Suppliers be requested to supply at least 6 months of consumables and/or reagents; that IPC be added as consideration; and that at a minimum request for information and details of equipment were confirmed prior to donations being shipped to avoid declining or disposal of medical equipment upon arrival in PICTs.

Session 8.3 National Medical Equipment Policy - 2

175. The second group to discuss the National Medical Equipment Policy noted how it can assist in managing equipment better and ensuring all health care facilities have the proper equipment. They believe it should outline the procurement process to better allow biomedical staff to engage/provide guidance, as well as outline details of regular preventative maintenance. The policy should also define scope – e.g. what is medical equipment – for biomedical staff to use it as a tool to push back on additional responsibilities that can be assigned to them (as discussed during the meeting). The policy should work to educate and improve decision making and provide overall guidelines for biomedical work.
176. They believe to progress the policy they need to influence management and discuss with stakeholders to involve them in the process and ensure they have the right information/understand the importance of the policy. They noted that Director level approval should be gained, and that activities should be mandated in business plans and an asset management system in place to support the work.
177. For changes and updated to the current generic policy template that noted the following additions should be included:
 - Those working on medical equipment should have technical training.
 - Visiting teams and consultants should be directly addressed.

- MEL component/framework should be included to ensure implementation is being tracked.
- Involvement of biomedical staff and expertise throughout the full process should be articulated; and
- Ability to decline donations and equipment should be accounted for.

9.0 Pacific Biomedical Engineering Network Terms of Reference

175. The group discussed all components of the Terms of Reference (ToR) for the Pacific Biomedical Engineering Network and made adjustments and updates by consensus. The name of the group was changed with the addition of “engineering” to differentiate from biomedical science. The updated ToR is provided at [Annex 2](#).

Closing

176. SPC’s Biomedical Technical Adviser Sunema Talapusi, provided a summary of discussions across the two days (a summary of which is provided at **Annex 1**) and highlighted key areas for ongoing engagement across the network including policy development, IPC, training and qualifications and biomedical waste disposal. The closing prayer was provided by Papua New Guinea.

Pacific Biomedical Network Meeting

Annex 1: Key Points and Recommendations

5th Pacific Biomedical Network Meeting

Key Points & Recommendations

Session 2:

Network

- Use of Regional Biomedical network to gain advocacy, advice and technical assistance in below areas:

Procurement

- Policies and procedures lead to challenges in obtaining budget and equipment in a timely and efficient manner for procurement of equipment, servicing and supplies.
- Biomedical staff and clinicians to collaborate on, and better inform, the planning and specifications for equipment for sustainable procurement.
- Service contracts and consumables for equipment should be incorporated into national budgets for sustainability.

Biomedical Waste Disposal

- Collaboration with stakeholders including government agencies (Environment and Finance); regional organisations (SPREP), and private sector (waste and transport companies) to ensure disposal is done in cost-effective ways that reduce carbon footprint and environmental impact.
- Need for legal frameworks and procurement policies which enforce greater responsibility of suppliers for ongoing servicing, support, and disposal (linked to procurement processes above).

Donation Policy

- Engagement and support from political leaders to ensure that donation policies and procedures are followed so that procurement and disposal aligns to needs of MoH.
- Biomed involvement should be from the beginning of the donation process and have priority list of medical equipment needs (standardization)
- Build-up of donated medical equipment causing workspace and storage issue in many PICT

Session 3

Capacity building:

- Increased professional development opportunities – regionalisation of training, education and capacity-building including development of regional plan.
- Importance of face-to-face **certified** training (like The Medical Room) and building capacity for maintenance and servicing at a national level, supported by SPC, development partners and training organisations.
- COVID-19 and Capacity-Building: Maximise on the online and remote support developed during COVID-19 pandemic to provide more consistent and timely

support to technicians in-country.

- Research Dissemination: Sharing of learnings and recommendations for information of other PICTs.

Session 4

Professional Development & On-going Training:

- Introduction of Fiji National University's Certificate in Biomedical Engineering (Level 4) – 2-year programme.
 - First year with 4 quarters of 8 weeks of teaching and the second year of industry attachment (either in Fiji or home country)
 - Suggestion for re-evaluation of theoretical aspects of the courses against the practical skills required in biomedical sector.
 - Recommendations for building the biomedical career pathway i.e., suggestion for offering Diploma level Biomedical Engineering Programme
 - Great support and interest shown by the Pacific into the Fiji National University's Certificate in Biomedical Engineering (Level 4) Programme.
- Documentation of Biomed Work for sustainability:
 - Asset registry and management systems as essential.
 - Continuous professional development within the biomed unit
- Acquiring bachelor's degree in electrical and application to biomed work back in PICT.
- Exchange partnership programs between PICTs for training, mentoring and workforce coverage.
- Attachment with Australian Hospitals
- Recommends to SPC and countries to explore the opportunity to draft legislations specifically for Biomedical Engineering and BMEQ pertaining to the training, development, career pathways, safeguard, protection, certification, and registration of the profession in the region and within country context.
- With WHO/SPC/DFAT and other partners, these will be of interest. Explores pooled procurement/leasing of like high-end BMEQ for Digital radiology, laboratory, endoscopy.

Session 5

Situational Analysis of Biomedical Services in the Pacific

Workforce:

- Increasing number of biomedical staff across the region
- 94% of local biomedical staff working in country, 4% Expats from PICT and only 2% of Expat (other – (Philippines & Australia).
- Higher representation of women (from 8% in 2014 to 17% in 2023)
- Younger workforce - (66% currently employed are <40 years)
- Major shift of workforce from no formal qualifications to certificate/diploma level qualifications
- Higher number of those with Bachelor's degree qualifications (increase to 62%)

Overall Biomedical Situation

- Overall gradual development of national policies and systems
- Dependent on availability and capacity of national staff

In 2023 - Largest attendance group with new countries attending from American Samoa, Niue & PNG (16 countries response but FSM couldn't attend because of the Typhoon)

Research

- Endorsement to publish regional biomedical data analyses and research.
- Results and findings of to be shared amongst the network and distributed more widely (including research publication and dissemination in forums where appropriate and beneficial), to better understand regional and country-specific status of biomedical services and apply learnings and recommendations that address the specific needs and challenges of the region.

Session 6

Biomedical Policies

- National Medical Equipment Policy & Donation Policy are important and PICT countries are encouraged to approve and implement these policies to further progress and assist their biomed unit.
- Many PICTs raised concerns that the delay of the endorsement and implementation is at administrators' level and request for SPC to assist in progressing it along.
- The two generic policies were discussed in group work and suggestions made will be incorporated into the final draft before dissemination for each country to use. One of these was to include IPC as it should be a key component in the biomed work.

Session 7

Maintenance & On-going Support

- Preventative maintenance is essential in prolonging the lifespan of the medical equipment. However, most of the PICTs do not have test equipment like the electrical safety analyzer. Recommend for countries to procure this essential test equipment.
- General consensus is that biomed is responsible for more work outside of the scope of their job description which delays the progression of biomed unit. Discussion of responsibilities lines of demarcations should be made with other stakeholders in the hospital for the biomed to solely focus on medical equipment related matters.

Session 8

Terms of Reference

- Name change– Pacific Biomedical Engineering Network
 - To differentiate from the biomedical scientist however, this network includes biomedical engineers and technicians.
- Discussed the responsibilities and agreed to have a meeting every 2 years.
- Further discussions will be made within the group to have a biomedical technical committee.

Pacific Biomedical Network Meeting

Annex 2: PBEN Terms of Reference

Terms of Reference

Pacific Biomedical Engineering Network (PBEN) Meeting

1. Introduction

The Pacific Biomedical Engineering Network forum operates under the auspices of the Directors of Clinical Services (DCS) and Pacific Heads of Health (HoH). Its role is to improve the coordination, collaboration and information sharing between country biomedical engineering divisions as well as regional development partners. The overall aim is to provide PICTs ownership of, and a means to provide inputs into, biomedical engineering services and workforce development in the Pacific region.

2. Role

The PBEN forum's role is to ensure the HoH are provided with clear guidance, advice and support from their DCS to enable them to make informed decisions on priority issues relating to biomedical engineering matters and the development of biomedical engineering services in the Pacific.

3. Operating Principles

Pacific Biomedical Engineers meet as a technical group, providing a governance oversight role for regional functions specifically related to biomedical engineering services and workforce improvement in the Pacific,

- Accountability: members are accountable individually to their respective DCS, and collectively to the HOH
- Efficiency: PBEN forum will ensure meetings, out of session processes, sub-committee arrangements and secretariat arrangements are efficient, with a well-structured agenda, good chairing, and early circulation of meeting papers and reports.

4. Meeting Chair

To ensure the biennial meeting conducts its business in an efficient and effective manner. It is proposed the following be adopted with regard to chairing the meeting.

a. Chairperson

The country hosting the Pacific Health Ministers Meeting chairs the PBEN meeting.

b. Vice Chairperson

The Deputy Chair will be the country hosting the last Pacific Health Ministers Meeting.

This will ensure that a strong link with the HoH meeting is maintained and continuity in sharing institutional knowledge from one outgoing to incumbent chairpersons.

5. Membership

Membership of the PBEN forum will consist of the biomedical engineers and technicians from PICTs. Development partners and stakeholders will be invited as observers.

6. Attendance

PICTs are encouraged to send the Heads of Department or a representative to the biennial PBEN meeting.

7. Responsibilities

The PBEN will:

- a. Make recommendations on technical and policy issues, and strategic directions for biomedical engineering services and workforce development in the region to the HoH,
- b. Provide and encourage a mechanism for networking between biomedical workforce from PICTs, development partners and service providers,
- c. Support efforts geared at pooling of resources and coordinating assistance to meet shared needs,
- d. Provide a mechanism for information sharing and transfer of best practises between countries,
- e. Review the progress of implementation and ensure full accountability to PICs review and approve work plans, budgets and reports.

8. Meeting Schedule

The PBEN forum will meet biennial, prior to the DCS. This is to enable the secretariat to prepare PBEN recommendations and other papers/presentations for the DCS meeting.

9. Secretariat arrangements

Secretariat services for the PBEN will be provided by the Pacific Community (SPC).

10. Review of Terms of Reference (TOR)

The PBEN forum shall retain the right to review and amend the TOR periodically.

Pacific Biomedical Network Meeting

Annex 3: Participant List

5th Pacific Biomedical Network Meeting

29th -31st May 2023, Nadi Fiji

LIST OF PARTICIPANTS, PARTNER AGENCIES, OBSERVERS AND SECRETARIAT

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